Step 1: Check for Missing & Null Values (Completeness)

Before doing any analysis, we will verify if there are any missing or blank values across our columns.

```sql
--Data Cleaning
--1. Check for Missing & Null Values (Completeness)
SELECT
    COUNT(*) AS total_rows,
    SUM(CASE WHEN Category IS NULL OR Category = '' THEN 1 ELSE 0 END) AS missing_category,
    SUM(CASE WHEN name IS NULL OR name = '' THEN 1 ELSE 0 END) AS missing_name,
    SUM(CASE WHEN mrp IS NULL THEN 1 ELSE 0 END) AS missing_mrp,
    SUM(CASE WHEN discountPercent IS NULL THEN 1 ELSE 0 END) AS missing_discount,
    SUM(CASE WHEN discountedSellingPrice IS NULL THEN 1 ELSE 0 END) AS missing_selling_price,
    SUM(CASE WHEN weightInGms IS NULL THEN 1 ELSE 0 END) AS missing_weight
FROM zepto;
--There is no missing and null values
```

Results | Messages

	total_rows	missing_category	missing_name	missing_mrp	missing_discount	missing_selling_price	missing_weight
1	3732	0	0	0	0	0	0

This means our dataset has **complete data coverage** with no blank or `NULL` values in any of those key fields.

Step 2: Identify Duplicate Records (Uniqueness)

Duplicates can skew our aggregations (like average price or total inventory count). Checking for exact row duplicates using a `GROUP BY` clause.

```sql
SELECT
    Category,
    name,
    mrp,
    discountPercent,
    weightInGms,
    COUNT(*) AS duplicate_count
FROM zepto
GROUP BY Category, name, mrp, discountPercent, weightInGms
HAVING COUNT(*) > 1;
```

150 %

Results | Messages

	Category	name	mrp	discountPercent	weightInGms	duplicate_count
1	Chocolates & Candies	Kitchens of India Dal Makhani	11000	30	285	2
2	Chocolates & Candies	Mother's Recipe Tamarind Paste	5000	2	200	2
3	Cooking Essentials	Everest Kesari Milk Masala	20500	0	50	2
4	Cooking Essentials	Mother's Recipe Tamarind Paste	5000	2	200	2
5	Health & Hygiene	Baidyanath Isabgol	11000	22	100	2
6	Home & Cleaning	Ariel Matic Liquid Detergent Front Load	25000	0	1000	2
7	Ice Cream & Desserts	Kitchens Of India Dal Makhani	11000	30	285	2
8	Ice Cream & Desserts	Mother's Recipe Tamarind Paste	5000	2	200	2
9	Munchies	Everest Kesari Milk Masala	20500	0	50	2
10	Munchies	Mother's Recipe Tamarind Paste	5000	2	200	2
11	Paan Corner	Listerine Cool Mint Mouthwash - Mild Taste	15000	10	250	2
12	Packaged Food	Kitchens Of India Dal Makhani	11000	30	285	2
13	Packaged Food	Mother's Recipe Tamarind Paste	5000	2	200	2
14	Personal Care	Listerine Cool Mint Mouthwash - Mild Taste	15000	10	250	2

It returned `duplicate_count = 2` for several items, but items like **Mother's Recipe Tamarind Paste** or **Kitchens of India Dal Makhani** appear across *multiple different categories* (e.g., Chocolates & Candies, Cooking Essentials, Ice Cream & Desserts, Packaged Food)

This means that the products got misclassified or assigned to incorrect categories.

We cleaned the data in three quick steps using SQL:

1. **Removed Invalid Values:** Removed items with zero prices or zero weights.
2. **Fixed Category Errors:** Removed products listed under the wrong categories (like *Dal Makhani* under *Ice Cream & Desserts*).
3. **Removed Duplicates:** Kept only one unique entry per product by keeping the record with the highest available stock

```
--removing the misclassified prodcucts

SELECT DISTINCT
    Category,
    name,
    mrp,
    discountPercent,
    availableQuantity,
    discountedSellingPrice,
    weightInGms,
    outOfStock,
    quantity
INTO zepto_cleaned
FROM zepto
WHERE mrp > 0
  AND weightInGms > 0
  AND name IS NOT NULL
  -- Keep only the logical/primary category for these specific overlapping items
  AND NOT (
      (Category = 'Munchies' AND name IN ('Everest Kesari Milk Masala', 'Mother''s
Recipe Tamarind Paste'))
      OR
      (Category = 'Packaged Food' AND name IN ('Mother''s Recipe Tamarind Paste',
'Kitchens Of India Dal Makhani'))
      OR
      (Category IN ('Chocolates & Candies', 'Ice Cream & Desserts'))
  );
  -- Removed all the misclassified products
  IF OBJECT_ID('zepto_cleaned', 'U') IS NOT NULL
    DROP TABLE zepto_cleaned;

WITH RankedProducts AS (
    SELECT
        Category,
        name,
        mrp,
        discountPercent,
        availableQuantity,
        discountedSellingPrice,
```

```
        weightInGms,
        outOfStock,
        quantity,
        ROW_NUMBER() OVER (
            PARTITION BY Category, name
            ORDER BY availableQuantity DESC
        ) AS rn
    FROM zepto
    WHERE mrp > 0
      AND weightInGms > 0
      AND name IS NOT NULL
      -- Exclude all cross-category misclassifications
      AND NOT (
          (Category IN ('Chocolates & Candies', 'Ice Cream & Desserts', 'Munchies',
'Packaged Food')
           AND name IN ('Everest Kesari Milk Masala', 'Mother''s Recipe Tamarind Paste',
'Kitchens Of India Dal Makhani'))
          OR
          (Category IN ('Personal Care', 'Paan Corner')
           AND name LIKE '%Listerine%')
      )
)
SELECT
    Category,
    name,
    mrp,
    discountPercent,
    availableQuantity,
    discountedSellingPrice,
    weightInGms,
    outOfStock,
    quantity
INTO zepto_cleaned
FROM RankedProducts
WHERE rn = 1;
```

```sql
--2. Identify Duplicate Records (Uniqueness)
SELECT
    Category,
    name,
    mrp,
    discountPercent,
    weightInGms,
    COUNT(*) AS duplicate_count
FROM zepto_cleaned
GROUP BY Category, name, mrp, discountPercent, weightInGms
HAVING COUNT(*) > 1;
```

113 %

Results | Messages

Category	name	mrp	discountPercent	weightInGms	duplicate_count

Step 3: Check for Logical Anomalies & Invalid Values (Validity)

```sql
-- Step 3: Check for Logical Anomalies & Invalid Values (Validity)
--prices, weights, and quantities should not be zero or negative

SELECT *
FROM zepto_cleaned
WHERE mrp <= 0
   OR discountedSellingPrice < 0
   OR weightInGms <= 0
   OR quantity < 0;
```

113 %

Results | Messages

Category	name	mrp	discountPercent	availableQuantity	discountedSellingPrice	weightInGms	outOfStock	quantity

```sql
--Verify Price and Discount Logic:
SELECT
    name,
    mrp,
    discountPercent,
    discountedSellingPrice,
    ROUND(mrp * (1 - discountPercent / 100.0), 2) AS calculated_sp
FROM zepto_cleaned
WHERE ABS(discountedSellingPrice - (mrp * (1 - discountPercent / 100.0))) > 1; -- allowing a small rounding margin
```

13 %

Results | Messages

	name	mrp	discountPercent	discountedSellingPrice	calculated_sp
1	Amul Butter - Pasteurised (School Pack)	5900	3	5700	5723.000000
2	Amul Cheese - Block	24500	2	24000	24010.000000
3	Amul Cheese - Pepper	8500	3	8200	8245.000000
4	Amul Cheese - Processed Slices	12200	1	12000	12078.000000
5	Amul Cheese Slices	36000	0	35900	36000.000000
6	Amul Cheese Spread Creamy	8500	5	8000	8075.000000
7	Amul Cheese Spread Spicy Garlic	8500	3	8200	8245.000000
8	Amul Delicious Fat Spread - Cholesterol Free	1800	5	1700	1710.000000
9	Amul Fresh Cream	19700	3	19100	19109.000000
10	Amul Frozen Malai Paneer	7500	2	7300	7350.000000

Prices in Paise: Prices are stored in paise (e.g., 5900 = ₹59.00).
Rounded Discounts: The discountPercent column stores rounded whole numbers (e.g., 3% instead of the actual 3.39%).
Calculation Difference: Recalculating price using the rounded 3% discount gives ₹57.23 instead of the actual listed selling price of ₹57.00 (5700).
Key Takeaway: discountedSellingPrice is the exact ground truth price, while discountPercent is just a rounded display figure.

Step 4: Standardize Text and Categorical Data

```
--4. Step 4: Standardize Text and Categorical Data (Consistency)
SELECT DISTINCT Category, COUNT(*) AS item_count
FROM zepto_cleaned
GROUP BY Category
ORDER BY Category;
```

113 %

Results | Messages

	Category	item_count
1	Beverages	121
2	Biscuits	133
3	Chocolates & Candies	362
4	Cooking Essentials	476
5	Dairy, Bread & Batter	121
6	Fruits & Vegetables	89
7	Health & Hygiene	89
8	Home & Cleaning	183
9	Ice Cream & Desserts	362
10	Meats, Fish & Eggs	59
11	Munchies	474
12	Paan Corner	302
13	Packaged Food	362
14	Personal Care	302

It confirms that category names are clean and standardized (e.g., no duplicate entries like

Data Analysis

Q1. Find the top 10 best-value products based on the discount percentage

```
-- Q1. Find the top 10 best-value products based on the discount percentage.
SELECT DISTINCT TOP 10 name, mrp, discountPercent
FROM zepto_cleaned
ORDER BY discountPercent DESC;
```

113 %

Results | Messages

	name	mrp	discountPercent
1	Dukes Waffy Chocolate Wafers	45	51
2	Dukes Waffy Orange Wafers	45	51
3	Dukes Waffy Strawberry Wafers	45	51
4	Ceres Foods Fish Mustard Instant Liquid Masala	220	50
5	Ceres Foods Laal Maas Instant Liquid Masala	220	50
6	Ceres Foods Nalli Nihari Instant Liquid Masala	220	50
7	Chef's Basket Durum Wheat Elbow Pasta	160	50
8	Chef's Basket Durum Wheat Fusilli Pasta	160	50
9	Chef's Basket Durum Wheat Penne Pasta	160	50
10	Dukes Waffy Chocolate Wafer Rolls	150	50

Top Discounts: The highest discount percentage in the dataset is 51% (held by Dukes Waffy Chocolate, Orange, and Strawberry Wafers, all priced at an MRP of ₹45).

Deep Discounts (50% Off): Several other items across cooking essentials and packaged foods (like Ceres Foods Instant Liquid Masalas, Chef's Basket Pastas, and Dukes Waffy Wafer Rolls) sit right behind at a flat 50% discount

Q2.What are the Products with High MRP but Out of Stock

```
--Q2.What are the Products with High MRP but Out of Stock

SELECT DISTINCT name, mrp
FROM zepto_cleaned
WHERE outOfStock = 1 AND mrp > 300
ORDER BY mrp DESC;
```

113 %

Results | Messages

	name	mrp
1	Patanjali Cow's Ghee	565
2	MamyPoko Pants Standard Diapers, Extra Large (12...	399
3	Aashirvaad Atta With Mutigrains	315
4	Everest Kashmiri Lal Chilli Powder	310

Top out of stock Items:

1. Patanjali Cow's Ghee is the most expensive out-of-stock item at **₹565**.
2. MamyPoko Pants Standard Diapers follow at **₹399**.
3. Aashirvaad Atta With Multigrains (**₹315**) and Everest Kashmiri Lal Chilli Powder (**₹310**) round out the list.

Q3.Calculate Estimated Revenue for each category

```sql
--Q3.Calculate Estimated Revenue for each category
SELECT category,
SUM(discountedSellingPrice * availableQuantity) AS total_revenue
FROM zepto_cleaned
GROUP BY category
ORDER BY total_revenue;
```

13 %

Results | Messages

	category	total_revenue
1	Fruits & Vegetables	10588
2	Meats, Fish & Eggs	20272
3	Biscuits	24253
4	Dairy, Bread & Batter	52811
5	Beverages	53015
6	Health & Hygiene	62093
7	Home & Cleaning	116585
8	Ice Cream & Desserts	213209
9	Packaged Food	213949
10	Chocolates & Candies	213949
11	Personal Care	243989
12	Paan Corner	245501
13	Munchies	321512
14	Cooking Essentials	323216

Top Revenue Drivers: Cooking Essentials (₹3,23,216) and Munchies (₹3,21,512) generate the highest potential inventory revenue by far, driven by large product catalogs and steady stock levels.

Mid-Tier Categories: Categories like Paan Corner, Personal Care, Chocolates & Candies, and Packaged Food all sit tightly in the ₹2.1L to ₹2.4L range.

Lower Revenue / Niche Categories: Fruits & Vegetables (₹10,588) and Meats, Fish & Eggs (₹20,272) show the lowest total stock valuation, which makes sense given they are fresh, high-turnover items with smaller total item counts and strict shelf lives.

Q4. Find top 10 products where MRP is greater than ₹500 and discount is less than 10%.

```sql
-- Q4. Find top 10 products where MRP is greater than ₹500 and discount is less than 10%.
SELECT DISTINCT Top 10 name, mrp, discountPercent
FROM zepto_cleaned
WHERE mrp > 500 AND discountPercent < 10
ORDER BY mrp DESC, discountPercent DESC;
```

113 %

Results | Messages

	name	mrp	discountPercent
1	Dhara Kachi Ghani Mustard Oil Jar	1250	8
2	Saffola Gold (Jar)	1240	0
3	Dhara Filtered Groundnut Oil (Jar)	1050	1
4	Fortune Rice Bran Health Oil (Jar)	1050	1
5	Fortune Soyabean Oil	1005	0
6	Fortune Sunlite Refined Sunflower (Jar)	925	0
7	Surf Excel Matic Powder Front Load	810	7
8	Surf Excel Matic Top Load	720	9
9	Pedigree Dog Food Adult Meat & Rice	660	7
10	Lizol Double Concentrate Disinfectant Floor Clean...	650	8

Dominated by Cooking Oils: Bulk cooking oils and jars (*Dhara*, *Saffola*, and *Fortune*) occupy the top spots for highest MRP (ranging from ₹925 to ₹1,250) with little to no promotional discounts (0% to 8%).

Household & Pet Essentials: Premium cleaning products and pet food—such as *Surf Excel Matic* (₹720–₹810) and *Pedigree Dog Food* (₹660)—also feature heavily in this high-value, low-discount bracket.

Q5. Identify the top 5 categories offering the highest average discount percentage.

```sql
-- Q5. Identify the top 5 categories offering the highest average discount percentage.
SELECT Top 5 category,
ROUND(AVG(discountPercent),2) AS avg_discount
FROM zepto_cleaned
GROUP BY category
ORDER BY avg_discount DESC;
```

113 %

Results | Messages

	category	avg_discount
1	Fruits & Vegetables	15
2	Meats, Fish & Eggs	11
3	Biscuits	8
4	Chocolates & Candies	8
5	Packaged Food	8

Leader by Far: Fruits & Vegetables offers the highest average discount percentage at 15%, acting as a major acquisition driver.

Fresh Protein Runner-Up: Meats, Fish & Eggs comes in second with an average discount of 11%.

Packaged & Snack Tier: Biscuits, Chocolates & Candies, and Packaged Food all cluster closely behind at an average of 8%.

Q7.Group the products into categories like Low, Medium, Bulk.

The weight classification analysis reveals that quick-commerce inventory is heavily skewed toward small pack sizes, with **91.18% of products (3,132 items)** and **90.92% of total stock (12,829 units)** weighing under 1 kg at an average MRP of ₹141.

In contrast, medium (1–5 kg) and bulk (≥ 5 kg) items represent only 7.54% and 1.28% of the catalog respectively, reflecting clear price scaling up to an average MRP of ₹617 for bulk goods.

This demonstrates an operational focus on lighter, fast-moving items optimized for rapid rider delivery and warehouse pick-and-pack efficiency, while heavy bulk goods (such as large oil canisters or grain bags) are kept as a niche offering.

```sql
--Q7.Group the products into categories like Low, Medium, Bulk.
SELECT
    CASE
        WHEN weightInGms < 1000 THEN 'Low (< 1kg)'
        WHEN weightInGms < 5000 THEN 'Medium (1kg - 5kg)'
        ELSE 'Bulk (>= 5kg)'
    END AS weight_category,
    COUNT(*) AS total_products,
    ROUND(AVG(mrp), 2) AS avg_mrp,
    SUM(availableQuantity) AS total_units_in_stock
FROM zepto_cleaned
GROUP BY
    CASE
        WHEN weightInGms < 1000 THEN 'Low (< 1kg)'
        WHEN weightInGms < 5000 THEN 'Medium (1kg - 5kg)'
        ELSE 'Bulk (>= 5kg)'
    END
ORDER BY total_products DESC;
```

13 %

Results | Messages

	weight_category	total_products	avg_mrp	total_units_in_stock
1	Low (< 1kg)	3132	141	12829
2	Medium (1kg - 5kg)	259	263	1049
3	Bulk (>= 5kg)	44	617	232

Q8.What is the Total Inventory Weight Per Category

- The Heavyweights: Cooking Essentials ($1,354,009g$ or ~1.35 tonnes) and Munchies ($1,353,109g$ or ~1.35 tonnes) are neck-and-neck at the top, accounting for the vast majority of physical inventory weight in the dark stores. This makes sense given bulk items like cooking oils, flours, rice, and large snack family-packs.

- Mid-Tier Volume: Ice Cream & Desserts, Packaged Food, and Chocolates & Candies form a solid middle cluster, ranging between ~458k to 461k grams.

- Light & Fresh Categories at the Bottom: Fruits & Vegetables ($79,794g$) and Meats, Fish & Eggs ($46,324g$) have the lowest total weight, aligning perfectly

with fresh inventory models that maintain lean stock to prevent spoilage and ensure rapid turnover.

```sql
--Q8.What is the Total Inventory Weight Per Category
SELECT category, SUM(CAST(weightInGms AS BIGINT) * availableQuantity) AS total_weight
FROM zepto_cleaned
GROUP BY category
ORDER BY total_weight DESC;
```

113 %

Results | Messages

	category	total_weight
1	Cooking Essentials	1354009
2	Munchies	1353109
3	Ice Cream & Desserts	460845
4	Packaged Food	458665
5	Chocolates & Candies	458665
6	Home & Cleaning	349688
7	Paan Corner	308058
8	Personal Care	306498
9	Health & Hygiene	138062
10	Beverages	137143
11	Dairy, Bread & Batter	135943
12	Biscuits	82066
13	Fruits & Vegetables	79794
14	Meats, Fish & Eggs	46324

Query 9. Stockout & Revenue Risk Analysis

stockout risk query highlights exactly where replenishment gaps impact potential revenue. Across all categories, there are 399 out-of-stock SKUs representing an immediate unfulfilled demand value of ₹34,648

```sql
--Q9.Stockout & Revenue Risk Analysis

SELECT
    category,
    COUNT(*) AS out_of_stock_products,
    SUM(discountedSellingPrice) AS lost_revenue_potential
FROM zepto_cleaned
WHERE outOfStock = 1
GROUP BY category
ORDER BY lost_revenue_potential DESC;
```

113 %

Results | Messages

	category	out_of_stock_products	lost_revenue_potential
1	Munchies	57	5264
2	Cooking Essentials	57	5232
3	Chocolates & Candies	42	3569
4	Ice Cream & Desserts	42	3569
5	Packaged Food	42	3569
6	Personal Care	18	2133
7	Paan Corner	18	2133
8	Meats, Fish & Eggs	10	1817
9	Dairy, Bread & Batter	24	1724
10	Beverages	24	1724
11	Home & Cleaning	16	1496
12	Biscuits	35	1264
13	Health & Hygiene	10	877
14	Fruits & Vegetables	4	277

Highest Revenue Impact (Top 2 Categories):

- **Munchies** (57 SKUs out of stock, **₹5,264** potential revenue) and **Cooking Essentials** (57 SKUs out of stock, **₹5,232** potential revenue) lead in revenue risk.
- Together, these two categories account for **30.29% of total lost revenue potential**.

The Top 5 Revenue Risk Tier:

- **Munchies**, **Cooking Essentials**, **Chocolates & Candies** (₹3,569), **Ice Cream & Desserts** (₹3,569), and **Packaged Food** (₹3,569) make up **61.20% of the total revenue at risk** (₹21,203 out of ₹34,648).

Highest Average Ticket Price per Stockout:

- **Meats, Fish & Eggs** has only 10 out-of-stock items, but carries the **highest average selling price per item (~₹181.70)**. Losing stock here hurts premium ticket size per order faster than high-volume low-cost items like **Biscuits** (~₹36.11 per stockout).

Lowest Risk / Efficient Stocking:

- **Fruits & Vegetables** has the lowest stockout count with only **4 SKUs out of stock** (₹277 lost revenue potential), demonstrating tight supply chain management for highly perishable items.

Query10: Price Tier Segmentation (Budget vs. Mid-Range vs. Premium)

```sql
--Q.11 Price Tier Segmentation (Budget vs. Mid-Range vs. Premium)

SELECT
    CASE
        WHEN mrp <= 100 THEN 'Budget (<= ₹100)'
        WHEN mrp <= 300 THEN 'Mid-Tier (₹101 - ₹300)'
        ELSE 'Premium (> ₹300)'
    END AS price_tier,
    COUNT(*) AS total_products,
    SUM(availableQuantity) AS total_stock
FROM zepto_cleaned
GROUP BY
    CASE
        WHEN mrp <= 100 THEN 'Budget (<= ₹100)'
        WHEN mrp <= 300 THEN 'Mid-Tier (₹101 - ₹300)'
        ELSE 'Premium (> ₹300)'
    END
ORDER BY total_products DESC;
```

113 %

Results | Messages

	price_tier	total_products	total_stock
1	Budget (<= ?100)	1589	5988
2	Mid-Tier (?101 - ?300)	1487	6545
3	Premium (> ?300)	359	1577

price tier breakdown shows that Zepto's inventory strategy focuses heavily on the **Budget** and **Mid-Tier** price points, which together represent **89.55% of all product SKUs** and **88.83% of total inventory stock**.